

From TileComm to a Typed Tile Edge

Profiling-driven abstractions for fused DeepSeek-R1 MoE inference on $8\times$ MI350X:
the decode **weight wall** (QuantTile) + the TP4-prefill **all-reduce** (tile-fused
reduce-scatter)

Subha

AMD · advised by Simran Arora & Muhammad Osama
 $8\times$ MI350X (gfx950, all-to-all XGMI) · DeepSeek-R1 MXFP4 · TP4 $\times$ DP2 + DP-attention + EP ·
HipKittens + IRIS

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The arc — profiling reshaped the plan

Last time: a fused DeepSeek-R1 MoE region ($1.56\times$ *prefill*) + a *TileComm* traffic-shaping idea. This time I profiled the two candidate communication abstractions *directly on 8×MI350X* — and the measurement changed the plan.

What died

Traffic-shaping. Reorder-only comm scheduling is capped at $1-4\%$ over a good round-robin on a fully-connected fabric; the on-node probe was issue-bound.

The two real levers

(1) *decode weight wall*
→ fewer weight bytes (MXFP4 / a QuantTile).
(2) *TP4 prefill all-reduce*
= 60–85% of GEMM+AR
→ tile-fused overlap, real $1.4-1.7\times$ ceiling.

The unifier

A *typed tile edge* (stage/retire): a tile whose *residency* and *format* are first-class. *QuantTile-v0* is the falsifiable first milestone.

The abstraction is the same; the profiling told me which lowering to build first — and it is not the comm one.

Recap — the fused region works, and it told me where to look

- ▶ *Prefill*: $1.56\times$ over the unfused MORI-dispatch $\rightarrow$ aiter-fmoe $\rightarrow$ MORI-combine baseline (1247 vs 1941 μ s), correctness-gated (RMS 0.018).
- ▶ We fuse the expert *region*: the gather *produces the GEMM's packed layout*, so *7 kernels collapse to 3* — no sort, no standalone quant, no scatter-combine. HipKittens compute + IRIS comm.
- ▶ *Decode is only marginal* ($\sim 2\text{--}3\%$, 516.5 vs 527 μ s at matched fp8) — because decode is a *weight-memory wall* (next). Fusion cannot move the dominant cost there.

Two regimes, two bottlenecks

Prefill is token-scaled compute followed by a *TP4 all-reduce*. *Decode* streams *all 32 local experts' weights* every step with *no* TP all-reduce (DP-attention).

So the levers are *different per regime* — which is exactly what the new profiling had to pin down before committing to an abstraction.

The pivot — traffic-shaping is a weak lever (measured, honest)

The seed: the combine's $2.4\times$ (934→386 μs) from *one* hand-written line —
`build_combine_pull(interleave=True)` round-robins destination cells across the 8 XGMI links
(sorted-by-dst piles concurrent blocks onto one link; round-robin spreads them). I tried to generalize it
into a demand-aware scheduler. It doesn't:

cost model (`tilesched.py`, calibrated to
934/386)

schedule	model μs	vs naive
<i>sorted</i> (naive default)	927.8	1.0 $\times$
round-robin (today)	388.2	$2.4\times$
<i>proportional</i> (byte-aware)	388.3	(= rr)

demand-aware beats round-robin by only $1\text{--}4\%$,
growing slowly with expert skew.

Why it's capped

On a *fully-connected* 8-GPU XGMI fabric, *bytes-per-link* are fixed by the demand matrix. Reordering *when* blocks fire only avoids the pathological sorted order — it *cannot beat the hottest-link floor*.

And the probe was inconclusive

The on-node XGMI probe was *issue-bound* (fire-and-forget stores timed *issue rate*: $\sim 4.7\text{ TB/s}$, $>10\times$ the fabric) and showed $\sim 1.05\times$. *Inconclusive, not a refutation* — not a lever to bet on.

So I stopped tuning the order and re-profiled the two candidate comm abstractions directly on-node.

Profiling methodology — exactly what was measured

- ▶ *Node*: the dedicated 8×MI350X (gfx950), container subha_probe (torch 2.10+rocm7.2, triton 3.6, IRIS editable). Config *TP4*, *bf16*.
- ▶ *Unfused reference* = *what production does today*: a full-256-CU *torch* matmul of each rank's *K-slice partial GEMM*, then a separate *RCCL all-reduce*. Two ops, not fused.
- ▶ *Timing*: *median-of-30* CUDA-event, then *MAX over the 4 ranks* (Simran's rule: long all-reduce times are usually profiling artifacts — use in-kernel / max-over-ranks).

Each shape is a real R1 op

- ▶ 8192,7168,18432 = R1 *down-projection* (prefill).
- ▶ 8192,7168,7168 = R1 *attention out-projection* (prefill).
- ▶ 1024/256,7168,18432 = small-*M* / decode-ish *M* (the same op at lower token count).

No synthetic shapes — every (M, N, K) is an all-reduce-preceding GEMM on the TP4 prefill path.

Question the profile answers: is the TP4 all-reduce cheap-relative-to-GEMM (then comm fusion is pointless) or does it dominate?

Result — the all-reduce DOMINATES the prefill GEMM+AR sequence

Unfused = torch K-slice GEMM *then* RCCL all-reduce. Overlap ceiling = serial/AR (a tile-fused reduce-scatter that hides the short GEMM under the long AR floors at $\max(\text{GEMM}, \text{AR}) \approx \text{AR}$).

GEMM (M, N, K) — R1 role	GEMM 256CU (ms)	RCCL AR (ms)	serial GEMM+AR (ms)	AR share	ceiling serial/AR
8192,7168,18432 — down-proj	0.565	1.148	1.641	70%	1.43×
8192,7168,7168 — attn out-proj	0.252	1.147	1.357	85%	1.18×
8192,4608,36864 — example	0.606	0.750	1.269	59%	1.69×
1024,7168,18432 — small prefill	0.093	0.176	0.262	67%	1.49×
256,7168,18432 — decode- M	0.053	0.073	0.121	60%	1.66×

GEMM 256CU = per-rank K-slice torch GEMM; RCCL AR = RCCL all-reduce; ceiling = serial/AR.

The reading

The all-reduce is *59–85% of the serial GEMM+AR* — 1.2–4.5× the GEMM itself. *Comm dominates the sequence*; it is *not* cheap-relative-to-GEMM.

The lever

A tile-fused reduce-scatter has a real *~1.4–1.7× prefill ceiling* — stronger than traffic-shaping, and it validates Simran: *no AMD engine fuses the all-reduce*.

But the shipped overlap substrate is far off — realizing A is a real build

The two IRIS examples that already fuse a GEMM with a collective, run at 8192,4608,36864 (`--gemm_sms 128`):

substrate	fused GEMM+AR	vs unfused 1.269 ms
unfused torch GEMM + RCCL all-reduce (ref)	1.269 ms	1.0×
ex.09 one-shot all-reduce (IRIS)	5.898 ms	<i>4.7× SLOWER</i>
ex.08 atomics all-reduce (IRIS)	358.8 ms	<i>280× SLOWER</i>

Why they lose

ex.09 is a *Triton streamK GEMM at 471 TFLOP/s* (vs torch's 1149 — 2.4× off) + a per-tile-barrier one-shot AR that reads-all-ranks; ex.08's per-element cross-rank atomic AR is catastrophic (7.76 TFLOP/s). *You cannot fuse your way out of a 2.4×-slow GEMM.*

What A actually needs

(1) a *HipKittens producer/consumer GEMM body* at $\sim$ torch rate whose epilogue emits tile transfers, and (2) an *in-kernel reduce-scatter matching RCCL bandwidth* (unproven on IRIS). Both substantial. And A is *PREFILL-only* — decode has no TP all-reduce.

The other regime — decode is weight-bound, with no all-reduce to fuse

- ▶ Under *DP-attention* the TP all-reduce is *gone at decode* — there is nothing for abstraction A to fuse.
- ▶ The expert GEMM streams *all 32 local experts' ~ 1.4 GB of fp8 weights every step* — a fixed *$\sim 176 \mu\text{s}$ HBM floor* at 8 TB/s, independent of token count. Gather+combine $\approx 18\%$ of the region.
- ▶ So *decode $\approx 80\%$ weight floor + 20% boundaries*. Any comm fusion caps decode at $\sim 1.1\text{--}1.25\times$.

A (comm) = prefill throughput; B (format) = decode latency. They attack different regimes — and B owns the serving denominator.

The only high-ceiling decode lever

Move *fewer weight bytes*. That is a *representation* lever, not a comm/overlap one — and it is where serving latency is actually set.

$\Rightarrow$ *MXFP4 weights* ($\frac{1}{4}$ the bf16 bytes) directly attack the 80% floor. This is abstraction *B (QuantTile)*, and it is the higher-serving-impact of the two.

MXFP4 — the format lever already ships $\sim 1.6\times$ on the decode GEMM

Setup (real MX checkpoint)

- ▶ Model `amd/DeepSeek-R1-MXFP4`: OCP *MXFP4 W4A4*, group 32, *E8M0* per-block scale along K .
- ▶ *Route-1 W4A16* decode GEMM
..`decode_mxfp4_sat`: store B *fp4* ($\frac{1}{4}$ bytes), pre-swizzle, unpack `fp4`→`bf16` with the `gfx950 hardware cvt_scalef32_pk_bf16_fp4`, then *bf16 MFMA*. Weight-only `fp4` — the right decode choice (weight-bound).

Measured (8×MI350, same node)

- ▶ Standalone GEMM: MXFP4 $\sim 1.6\times$ over our `fp8_sat` decode, $\sim 2\times$ over `bf16` (RMS-vs-fp4-dequant 0.0033).
- ▶ Fused decode region (TOTAL_M=512): $520.6\mu s = 1.11\times$ over *fp8*, $1.69\times$ over *bf16* (region diluted by shared A-dequant + gather/act/combine).

Honest caveat + the hardware is ready

Baseline: our captured `aiter a4w4` EP kernel was *untuned and buggy* for our shape (every `block_size_M` sweep fell back to 2stage default) — so beating it means nothing. The real baseline is the reproducible *SGLang R1-FP4 e2e* (Simran offered to run it — the gold denominator). *HK's scaled-MFMA already supports fp4* (`mma_AbT_scaled`, `cbsz/blgp=4`) — a true `fp4`×`fp4` path is headroom, not a blocker.

The abstraction — a typed tile edge (`stage` / `retire`)

A tile whose *RESIDENCY* (may live on another rank) and *FORMAT* (may be compressed) are *first-class type parameters*. The author declares an edge; the compiler lowers each axis to its own backend.

```
// inbound: fill an HK tile (QuantTile = FORMAT)
stage(b_tile)
  .from( local() ) // RESIDENCY
  .format( mx_fp4{scale,32,E8M0} ) // FORMAT
  .for_mma<Bt>();

// outbound: dispose a tile (Overlap-A)
retire(c)
  .reduce_scatter( tp_group, sum ) // RESIDENCY
  .overlap_with( next ); // SCHEDULE
```

Two lowerings, one declaration

QuantTile (FORMAT, decode) → CDNA4 scaled-MFMA / in-register dequant. *Tile-fused reduce-scatter* (RESIDENCY, prefill) → IRIS RMA + a tile schedule. Same two verbs; the compiler dispatches on the type axes.

Honest boundary

An *API-level unifier, not one cost model*. The axes are orthogonal (residency $\neq$ numerics, format $\neq$ topology), so each keeps its *own* validated model — by design. No scalar objective spans μ s-of-comm vs HBM-bytes vs RMS.

QuantTile-v0 — the falsifiable first milestone

One HipKittens grouped-decode GEMM body serving *both* fp8-sat *and* MXFP4 through a compile-time tile descriptor — *no per-format kernel fork* (today they are a 90%-identical copy-paste; HK itself hard-codes one GEMM per format).

gate	threshold	anchor (measured, same-node)
<i>G1</i> fp8 no-regression	within $\pm 5\%$ of standalone	3.97 TB/s (fp8 _sat)
<i>G2</i> mxfp4 win preserved	$\geq 1.5\times$ over fp8 (5% slack on $1.6\times$)	$1.6\times$
<i>G3</i> correctness	fp8 / mxfp4 RMS $\leq 5e-3$ vs faithful dequant	0.0037 / 0.0033
<i>G4</i> region still wins	$\geq 1.10\times$ (fused decode, TOTAL_M=512)	520.6 vs 578.9 μ s

Why this first

Highest serving impact (attacks the *dominant decode weight wall*) + cleanest novelty (the *format/lowering* axis). Falsifiable against numbers we already trust — *not* against a captured baseline.

Why not comm first

The overlap/comm axis is *crowded* (Flux, CoCoNet, MSCCL++, TRT-LLM) *and* needs a new GEMM body before it can even match unfused (slide 6). Format-first is the cheaper, more differentiated proof.

Roadmap & asks

1. *QuantTile-v0* — one HK decode body, fp8 + MXFP4 via a tile descriptor, gated G1–G4. The falsifiable proof the FORMAT axis is real.
2. *Tile-fused reduce-scatter* for the TP4 prefill AR — the RESIDENCY axis; the schedule is Osama's "how many tiles before you reduce"
($G^* \approx \sqrt{PL/(t_c + t_m)}$), first target: *match* unfused, then claw toward the $1.4\text{--}1.7\times$ ceiling.
3. *e2e SGLang R1-MXFP4 profile* under TP4×DP2 + DP-attention + EP — the gold denominator that decides the real baseline.

What I'd like from you

- ▶ *Simran* — the e2e SGLang R1-FP4 run (gold baseline) + a per-expert token-count trace.
- ▶ *Osama* — gut-check the tile-fused reduce-scatter schedule (the *G* sweep) as the overlap target.
- ▶ *Awad* — is stage / retire the right *shape* for the tile edge?

Honest limits: traffic-shaping is weak; the overlap substrate is far off ($4.7\text{--}280\times$ today); A is prefill-only; decode carries an fp8 RMS floor; the MXFP4 baseline is not yet external.

Backup — the empty quadrant (prior art, adversarially searched)

The claim: *a typed tile whose remote residency AND quantized representation are both first-class*, co-lowered to IRIS device-RMA + CDNA4 scaled-MFMA — compression kept live *from HBM across XGMI into the MFMA accumulate*.

	no residency axis	residency first-class
no format axis	HipKittens (tile compute, no comm)	IRIS / Triton-distributed (opaque dtypes)
format first-class	Tilus (single-GPU, no transport)	<i>this work (stage/retire)</i>

- ▶ *Overlap line* (Flux, CoCoNet, MSCCL++, TRT-LLM fused AR+RMSNorm+quant) owns residency; data crosses in native dtype or dequants *before* the matmul / *after* the reduce.
- ▶ *Low-precision line* (Tilus, LiquidGEMM) owns representation but is single-GPU — no transport boundary to preserve compression across.
- ▶ *Nearest threat: DeepGEMM “Mega MoE” (DeepSeek-V4)* — the mechanism twin, but a *hand-written Blackwell mega-kernel*, not an abstraction, NVLink not IRIS, MoE-combine not the TP all-reduce. We distinguish on *programming-model + AMD (IRIS+CDNA4)* — never on “first to send fp4 over the wire” (MORI/DeepEP already do).

Backup — the honest ledger (measured / claimed / unknown)

item	status
FORMAT lever real (decode weight wall)	<i>MEASURED</i> — MXFP4 $\sim 1.6\times$ over fp8; region 1.11/1.69 $\times$
fp8 _sat decode = 3.97 TB/s (G1 anchor)	<i>FLAGGED</i> — re-verify on the HK build before v0 rests on it
RESIDENCY lever real (TP4 AR = 59–85%)	<i>MEASURED</i> — slide 5
Overlap-A ~ 1.4 – $1.7\times$ prefill ceiling	<i>CEILING</i> — $\max(\text{GEMM}, \text{AR})$, not a fused run
A on the current IRIS substrate	<i>NO</i> — examples 4.7–280 $\times$ slow; needs new GEMM body + RS
compositional cell (fp4-over-XGMI combine)	<i>UNMEASURED</i> — the one cell the split models don't cover
one unified cost model over the 5 currencies	<i>DOES NOT EXIST</i> — and the thesis must not claim it
traffic-shaping as the headline lever	<i>REJECTED</i> — 1–4% over round-robin; probe issue-bound

RCCL bus BW here ≈ 150 GB/s (modest — under-tuned for these sizes on gfx950, or per-iter barrier overhead). The robust takeaway is the *ratio* (AR share), not the absolute BW.

Backup — anticipated questions

- ▶ *“Why abandon the comm abstraction Awad asked for?”* I didn’t — I re-pointed it. Traffic-shaping (the *scheduling* half) is capped at 1–4%. The comm lever that *is* real is *fusion* (Overlap-A, tile-fused reduce-scatter) on the TP4 prefill AR — kept as the second lowering of the same stage/retire notation, with a measured $1.4\text{--}1.7\times$ ceiling and an honest build cost.
- ▶ *“Isn’t ‘residency + format’ just one big cost model you don’t have?”* Correct that we don’t have it — and we deliberately don’t claim it. The axes are *orthogonal* (residency doesn’t change numerics; format doesn’t change topology), so each lowers to its *own* validated model. The word “unify” applies to the *declaration surface*, never to the objective function.
- ▶ *“Did you get a new end-to-end MXFP4 speedup?”* No — deliberately. Route-1 is shipped ($\sim 1.6\times$ GEMM, $1.11\times$ region); the missing piece is a *legitimate external baseline* (the SGLang R1-FP4 e2e), because the a4w4 kernel we captured was untuned/buggy. Never iterate against a kernel we captured ourselves.
- ▶ *“Why QuantTile before the combine or Overlap-A?”* It attacks the *dominant* decode cost (the 80% weight floor), it is falsifiable against trusted numbers (G1–G4), and it needs *no* new GEMM body — the two fast decode kernels already exist; v0 makes them *one* at zero cost.